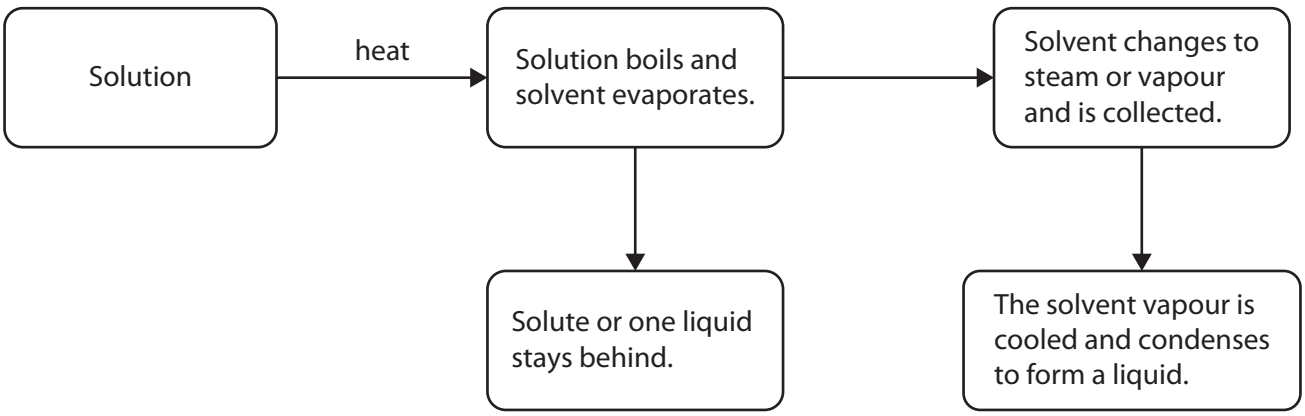
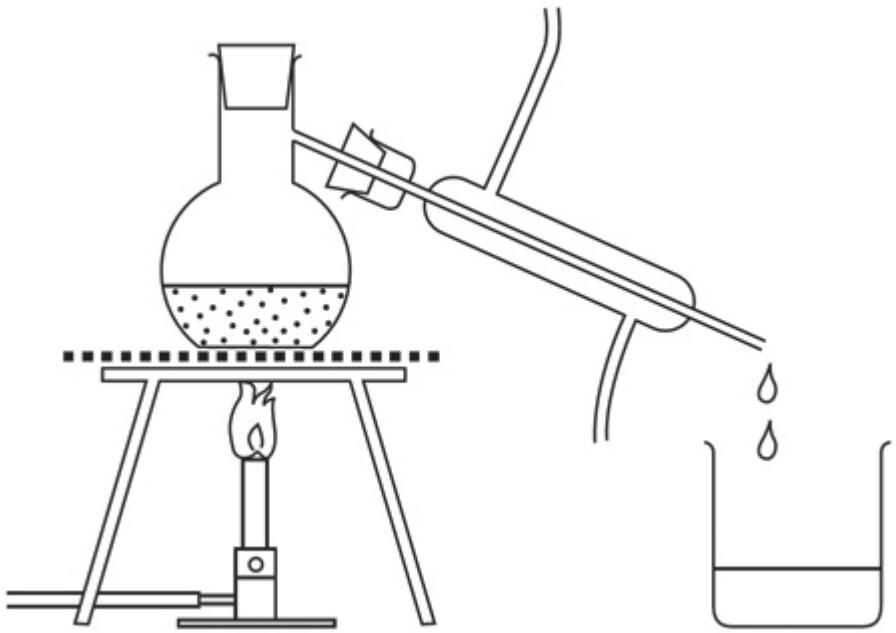
# Distillation (1)

**Distillation** is used to separate dissolved substances when you want to keep both the liquid and the solute. Distillation can also be used to separate a liquid from another liquid. This is because different liquids have different boiling points. The process of distillation uses two changes of state, **evaporation** and **condensation**. The material collected during distillation is called the **distillate**.

The flowchart below shows the steps involved in **distillation**.




Distillation is used to make alcoholic spirits, to produce oils and perfumes from plants, to make distilled water and to refine oil. In a science laboratory a **Leibig's Condenser** can be used for distillation. The condenser uses cold water in a tube to cool the evaporated liquid.



When set up, a Leibig's Condenser looks like this.

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# Distillation (2)

Read the Resource sheet 'Distillation (1)', then answer the following questions.

☞ **1** How is distillation different to evaporation?

_____

**2** Which two changes of state are involved in distillation?

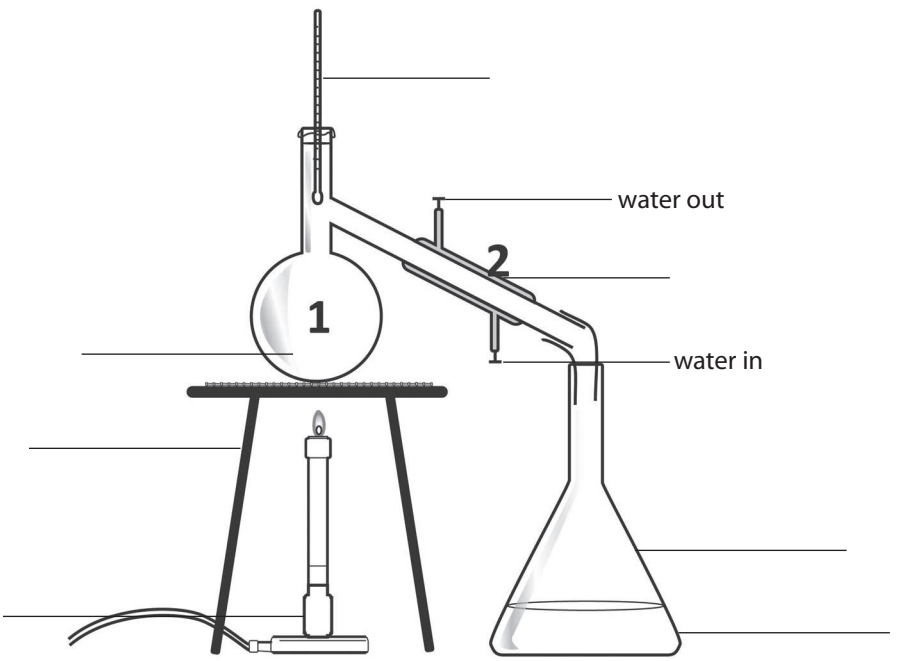
_____

**3** Jeremy wants to separate and collect the solvent from a salt solution using the process of distillation. Number the steps he needs to take in the correct order.

_____ Gently heat the salt solution in the conical flask until nearly all the water has evaporated.

_____ Set up equipment as shown in diagram below.

_____ Connect tubing to Leibig's Condenser and turn on water. Run water through condenser.




**Word bank**

salt water
thermometer
tripod
conical flask
Bunsen burner
pure water
condenser

**4** Use the words in the word bank to label the diagram.

**5** Name the change of state that occurs at:

1 _____

2 _____


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